

Probability distributions

Basic Terminology and Definitions

In probability theory, an *experiment (trial)* is any procedure that can be repeated infinitely many times and has a well-defined set of possible outcomes, known as the *sample space*, usually denoted by S . An experiment is said to be *deterministic* if a unique input gives rise to a unique output. An experiment is said to be *random* if when repeated in exactly the same way, it can give rise to more than one possible outcome. A random experiment that has exactly two distinct possible outcomes is known as a *Bernoulli trial*. The two distinct outcomes are referred to as *success* and *failure*.

Thus, in a random experiment, since the occurrence of outcomes are not certain, we can only talk about the likelihood of a certain outcome. The measure of the likelihood of an occurrence is called the *probability* of that occurrence (*event*). Probability of an event A , in fact, is a number that satisfies the following axioms:

- i. $P(S) = 1$
- ii. $0 \leq P(A) \leq 1$
- iii. $P(A \cup B) = P(A) + P(B)$; where the events A and B are mutually exclusive.

A *random variable* is a function that assigns a real number to each outcome in the sample space of an experiment. Random variables are denoted by capital letters $X, Y, \dots$ and their possible values are denoted by the corresponding small case letters $x, y, \dots$. A *discrete* random variable has finite or countably infinite range, and *continuous* random variable has an interval of real numbers for its range.

A *probability distribution* is a mathematical function that provides the probabilities of different possible outcomes in a random experiment. Probability distributions are generally divided into two classes: *discrete* and *continuous*.

Discrete Probability Distributions

A *discrete probability distribution* is associated with experiments whose set of possible outcomes is discrete, such as a coin toss or a roll of a die. It is a discrete list of the probabilities of the outcomes, known as a *probability mass function* (pmf). We will denote this function as $f(x)$. This function is specified by assigning a probability to each element x of the sample space.

Example: Toss three coins

Properties of Probability Mass Functions

Let X be a discrete random variable with values $\{x_1, x_2, \dots, x_n\}$

Then,

- i. For all $j = 1, \dots, n$, $f(x_j) \geq 0$
- ii. $\sum_{j=1}^n f(x_j) = 1$

Verify for the example. Draw a graph.

The cumulative distribution function

The cumulative distribution function, denoted $F(x)$ is defined as

$$F(x) = \sum_{x_j \leq x} f(x_j)$$

Example

It has some important properties:

- i. $\lim_{x \rightarrow -\infty} F(x) = 0$
- ii. $\lim_{x \rightarrow \infty} F(x) = 1$
- iii. F is bounded between 0 and 1
- iv. F is non-decreasing
- v. F is over all $\Re$

Thus, a discrete random variable can be defined as a random variable whose cumulative distribution function increases only by jump discontinuities—that is, its cdf increases only where it "jumps" to a higher value, and is constant between those jumps. The points where jumps occur are precisely the values which the random variable may take.

Expected value

The expected value of a discrete random variable is denoted as $E(X)$ or μ . We have

$$E(X) = \mu = \sum_{j=1}^n x_j f(x_j)$$

Thus, this is a weighted average of possible values using where the weights are their probabilities.

The expected value can be any real number.

Variance

Variance is an important measure of the dispersion of the distribution. It is denoted as $V(X)$. We have

$$V(X) = \sum_{j=1}^n (x_j - \mu)^2 f(x_j) = \sum_{j=1}^n x_j^2 f(x_j) - \mu^2$$

The *standard deviation*, σ , is the square root of the variance, and hence another measure of dispersion.

The variance (and the standard deviation) can be any positive real number.

Well-known discrete probability distributions used in statistical modeling include the *binomial distribution*, the *geometric distribution*, the *negative binomial distribution*, the *hypergeometric distribution*, and the *Poisson distribution*.

The Binomial Distribution

Suppose we have a series of Bernoulli trials (i.e., two possible outcomes called success and failure) such that

- i. The number of trials n is fixed
- ii. The trials are independent
- iii. The probability of success, p , remains a constant (selections with replacement).

In that case, the random variable X denotes the number of successes and the probability of x successes (pmf) is given as

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

for $x = 0, 1, 2, \dots, n$

The expected value is

$$E(X) = \mu = np$$

and the variance is

$$V(X) = np(1-p)$$

The Geometric Distribution

Suppose we have a series of Bernoulli trials (i.e., two possible outcomes called success and failure) such that

- i. The trials are independent
- ii. The probability of success, p , remains a constant (selections with replacement).

In that case, the random variable X denotes the number of trials needed until the first success and the pmf is

$$f(x) = (1-p)^{x-1}p$$

for $x = 1, 2, \dots$

The expected value is

$$E(X) = \mu = \frac{1}{p}$$

and the variance is

$$V(X) = \frac{1-p}{p^2}$$

The Negative Binomial Distribution

Suppose we have a series of Bernoulli trials (i.e., two possible outcomes called success and failure) such that

- i. The trials are independent
- ii. The probability of success, p , remains a constant (selections with replacements).

In that case, the random variable X denotes the number of trials needed until the r th success and the pmf is

$$f(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r$$

for $x = r, r+1, r+2, \dots$

The expected value is

$$E(X) = \mu = \frac{r}{p}$$

and the variance is

$$V(X) = \frac{r(1-p)}{p^2}$$

The Hypergeometric Distribution

Suppose in a set of N objects K can be classified as successes and $(N - K)$ as failures. A sample of n objects is selected randomly *without replacement*. The random variable X equals the number of successes in the sample and the pmf is

$$f(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

The expected value is given as

$$E(X) = \mu = n \frac{K}{N}$$

and the variance is

$$V(X) = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \left(\frac{N-n}{N-1}\right)$$

The Poisson Distribution

Suppose we have events that occur regularly in a unit interval (time, length, etc.) λ many times on the average. The random variable X denotes the number of occurrences in a unit interval. Then, the pmf is given as

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

for $x = 1, 2, \dots$

The expected value is

$$E(X) = \mu = \lambda$$

and the variance is

$$V(X) = \lambda$$

Continuous Probability Distributions

A *continuous probability distribution* is associated with experiments whose set of possible outcomes is an interval in the real line such as time, length, pressure, etc. In this case, the probability distribution is a continuous function known as a *probability density function* (pdf). We will denote this function as $f(x)$.

Properties of Probability Density Functions

Then,

i. For all $x, f(x) \geq 0$

ii. $\int_{-\infty}^{\infty} f(x)dx = 1$

When a random variable takes values from a continuum then typically, only events that include infinitely many outcomes, such as intervals, can have positive probability. We calculate probabilities as follows:

$$P(X < a) = \int_{-\infty}^a f(x)dx$$

$$P(X > a) = \int_a^{\infty} f(x)dx$$

$$P(a < X < b) = \int_a^b f(x)dx$$

Note that, the probability for X to take any single value a (that is $a \leq X \leq a$) is zero, because an integral with coinciding upper and lower limits is always equal to zero.

Any one of these can have equality.

The cumulative distribution function

The cumulative distribution function, denoted $F(x)$ is defined as

$$F(x) = \int_{-\infty}^x f(t)dt$$

Example and graph

It has some important properties:

- i. $\lim_{x \rightarrow -\infty} F(x) = 0$
- ii. $\lim_{x \rightarrow \infty} F(x) = 1$
- iii. F is bounded between 0 and 1
- iv. F is non-decreasing
- v. F is over all $\Re$

A continuous probability distribution is a probability distribution that has a cumulative distribution function that is continuous.

Expected value

The expected value of a continuous random variable is denoted as $E(X)$ or μ . We have

$$E(X) = \mu = \int_{-\infty}^{\infty} xf(x)dx$$

Thus, this is a weighted average of possible values using where the weights are their probabilities.

The expected value can be any real number.

Variance

Variance is an important measure of the dispersion of the distribution. It is denoted as $V(X)$. We have

$$V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

The *standard deviation*, σ , is the square root of the variance, and hence another measure of dispersion.

The variance (and the standard deviation) can be any positive real number.

Well-known continuous probability distributions used in statistical modeling include the *uniform distribution*, the *normal distribution*, the *exponential distribution*, the *gamma distribution*, the *chi-squared distribution*, *Student's t-distribution*, and the *F* distribution

The Normal Distribution

The most commonly used model for a continuous measurement is the *normal random variable*. A random variable X with probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

is a normal random variable with

$$E(X) = \mu$$

and

$$V(X) = \sigma^2$$

Here, $-\infty < x < \infty$.

A normal distribution with expected value μ and standard deviation σ (and hence variance σ^2) is denoted as $N(\mu, \sigma^2)$.

A very important special case is the *standard normal variable* denoted Z . This distribution has $\mu = 0$ and $\sigma = 1$. So, the pdf is

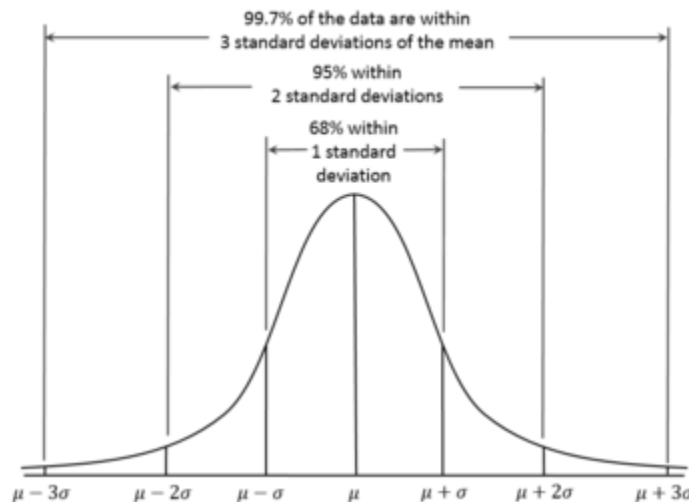
$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Properties of the Normal Distribution

1. The probability density function is bell-shaped.
2. It reaches its maximum at $x = \mu$.
3. It is symmetric with respect to the line $x = \mu$.
4. The graph is always above the horizontal axis. As $x \rightarrow \pm\infty$, the curve gets closer and closer to the horizontal axis without ever touching it (i.e., the x -axis is the horizontal asymptote).
5. Clearly, the total area under the curve is 1. Thus, by virtue of symmetry, the area to the left of $x = \mu$ = the area to the right of $x = \mu = \frac{1}{2}$.

6. Changing the value of μ shifts the graph to the left or to the right without changing the overall shape.
7. The standard deviation σ is the distance between the mean and the inflection point. Thus, changing the value of σ does affect the shape of the curve. A small σ implies a tall and narrow curve and a large σ implies a short and wide curve.
8. **The Empirical Rule** (or the *3-sigma rule*)
 - (i) The area under the curve between $x = \mu - \sigma$ and $x = \mu + \sigma$ is approximately 0.68
 - (ii) The area under the curve between $x = \mu - 2\sigma$ and $x = \mu + 2\sigma$ is approximately 0.95
 - (iii) The area under the curve between $x = \mu - 3\sigma$ and $x = \mu + 3\sigma$ is approximately 0.997

Thus, in a normally distributed data set it is unusual to observe values that are more than two standard deviations away from the mean.



9. Any normal distribution can be standardized by the transformation

$$z = \frac{x - \mu}{\sigma}$$

that is,

$$\frac{\text{observed value} - \text{mean}}{\text{standard deviation}}$$

10. Conversely, we can transform $N(0,1)$ to $N(\mu, \sigma^2)$ by the transformation

$$x = \mu + \sigma z$$

The Exponential Distribution

The random variable X that equals the distance (time) between successive events from a Poisson process with mean number of events λ per unit interval is called an *exponential random variable* with parameter λ . The probability density function is

$$f(x) = \lambda e^{-\lambda x}$$

for $x > 0$.

The expected value is

$$E(X) = \mu = \frac{1}{\lambda}$$

and the variance is

$$V(X) = \frac{1}{\lambda^2}$$

The Gamma Distribution

Recall that the gamma function is defined as

$$\Gamma(r) = \int_0^{\infty} x^{r-1} e^{-x} dx$$

for $r > 0$. It can be shown that

$$\Gamma(1) = 1$$

and

$$\Gamma(r) = (r-1)\Gamma(r-1)$$

This shows that for any positive integer value of r ,

$$\Gamma(r) = (r-1)!$$

It can also be shown that

$$\Gamma(1/2) = \sqrt{\pi}$$

Thus, we can compute values of the gamma function at positive integers and halves of positive integers.

An exponential distribution describes the length of time until the first event is obtained in a Poisson process with average number of occurrences λ . If, instead, we are interested in the r th event, then the random variable is called a ***gamma random variable***. The pdf is given as

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}$$

for $x > 0$.

Many distributions are special cases of the gamma distribution. Of course, the exponential distribution is a special case of the gamma distribution with $r = 1$. If r is an integer, the distribution is called the ***Erlang distribution***. If $\lambda = \frac{1}{2}$ and $r = 1 + \frac{n}{2}$, we get a ***chi-squared distribution*** with parameter n .

The parameter λ is called the ***scale parameter*** and the parameter r is called the ***shape parameter***.

The expected value is

$$E(X) = \mu = \frac{r}{\lambda}$$

and the variance is

$$V(X) = \frac{r}{\lambda^2}$$

The Chi-Squared Distribution

The **chi-squared distribution** (χ^2 -distribution) with n degrees of freedom is the distribution of a sum of the squares of n independent standard normal random variables. In other words, if $Z_1, Z_2, \dots, Z_n$ are independent, standard normal random variables, then the sum of their squares,

$$Z_1^2 + Z_2^2 + \dots + Z_n^2 = \sum_{j=1}^n Z_j^2$$

has a chi-squared distribution.

The chi-squared distribution is a special case of the gamma distribution and is one of the most widely used probability distributions in inferential statistics, notably in hypothesis testing or in construction of confidence intervals. The primary reason that the chi-squared distribution is used extensively in hypothesis testing is its relationship to the normal distribution.

The probability density function (pdf) of the chi-square distribution is

$$f(x) = \frac{x^{n/2} e^{-x/2}}{2^{n/2} \Gamma(n/2)}$$

for $x > 0$.

The expected value of the chi-squared distribution with n degrees of freedom is

$$E(X) = \mu = n$$

and the variance is

$$V(X) = 2n$$

DISCRETE PROBABILITY DISTRIBUTIONS USING RSTUDIO

The Binomial Distribution

To generate k random numbers from a binomial distribution with number of trials N and probability of success p_s , we write

```
> rbinom (n = k, size = N, prob = p_s)
```

So, for example, if we want three random numbers from a binomial distribution with total number of trials 15, and probability of success 0.4, we write

```
> rbinom (n = 3, size = 15, prob = 0.4)
```

```
[1] 6 4 5
```

To get the value of the pmf at $X = j$, $P(X = j)$, we write

```
> dbinom (x = j, size = N, prob = p_s)
```

So, for example,

```
> dbinom (x = 4, size = 15, prob = 0.4)
```

```
[1] 0.1267758
```

To get $P(X = k)$, $P(X = k + 1)$, ..., $P(X = m)$, we write

```
> dbinom (x = k:m, size = N, prob = p_s)
```

So, to construct a table of binomial probabilities we write

```
> dbinom (x = 0:N, size = N, prob = p_s)
```

For example, to get a table of binomial probabilities with number of trials 5 and probability of success 0.3, we write

```
> dbinom (x=0:5, size = 5, prob = 0.3)
```

and get

```
[1] 0.16807 0.36015 0.30870 0.13230 0.02835 0.00243
```

To write this output in a tabulated format, we use the **cbind** (column bind) function and write

```
> w <- dbinom (x = 0:N, size = N, prob = p_s)
```

```
> cbind(0:N, w)
```

To get the value of the cdf at k , $P(X \leq k)$,

```
> pbinom (q = k, size = N, prob = p_s)
```

To graph a binomial distribution as a discrete set of points, we write:

```
>x <- 0:N
>y <- dbinom(x, size = N, prob = p_s)
>plot(x, y)
```

To get quartiles, we write

qbinom (p = the given left-hand side probability, $\text{size} = N$, $\text{prob} = p_s$)

So, for instance to find the x -value that has an area of 0.66 to its left in a binomial distribution with 12 trials and probability of success 0.41, we write

```
>qbinom(p = 0.66, size = 12, prob = 0.41)
[1] 6
```

The Geometric Distribution

To get k random numbers from a geometric distribution with probability of success p_s , we write

```
>rgeom(k, p_s)
```

Note that these random numbers represent the number of failures before the first success.

To get the pmf at $X = a$, $P(X = a)$, we write

```
>dgeom(a - 1, p_s)
```

We can also write

```
>dgeom(a - 1:b - 1, p) to get  $P(X = a), P(X = a + 1), \dots, P(X = b)$ 
```

To write this output in a tabulated format, we use the **cbind** (column bind) function and write

```
>w <- dgeom(a - 1:b - 1, p)
>cbind(0:N, w)
```

To get the value of the cdf at a , $P(X \leq a)$, we write

```
>pgeom(a - 1, p_s)
```

To graph a geometric distribution as a discrete set of points, we write:

```
>x <- 0:n
```



```
>y <- dgeom(x, p_s)
>plot(x, y)
```

To get quartiles, we write

```
>qgeom(p = the given left-hand side probability, p_s)
```

So, for instance to find the x -value that has an area of 0.75 to its left in a geometric distribution with probability of success 0.47, we write

```
>qgeom(p = 0.75, 0.47)
[1] 2
```

The Negative Binomial Distribution

To generate k random numbers from a negative binomial distribution with r successes and probability of success p_s , we write

```
> rnbinom(n = k, size = r, prob = p_s)
```

Note that these random numbers represent the number of failures before the r th success.

To get the value of the pmf, we write

```
>dnbinom(x = number of trials - r, size = r, prob = p_s)
```

To get $P(X = k), P(X = k + 1), \dots, P(X = m)$, we write

```
>dnbinom(x = k - r:m - r, size = r, prob = p_s)
```

So, to construct a table of binomial probabilities we write

```
>dnbinom(x = 0:N, size = r, prob = p_s)
```

To write this output in a tabulated format, we use the **cbind** (column bind) function and write

```
>w <- dnbinom(x = 0:N, size = r, prob = p_s)
>cbind(0:N, w)
```

To get the value of the cdf,

```
>pnbinom(q = number of trials - r, size = r, prob = p_s)
```

To graph a negative binomial distribution as a discrete set of points, we write:

```
>x <- 0:n  
>y <- dnbinom(x, size = r, prob = p_s)  
>plot(x, y)
```

To get quartiles, we write

qnbinom (p = the given left-hand side probability, size = r , prob = p_s)

The Hypergeometric Distribution

Suppose we have the following:

m is the number of successes in the population

n is the number of failures in the population

x is the number of successes in the sample

q is the number of failures in the sample

k is the sample size

To generate a random numbers from a hypergeometric distribution with the above parameters, we write

```
> rhyper(nn = a, m = , n = , k = )
```

To get the value of the pmf, we write

```
> dhyper(x = , m = , n = , k = )
```

To get the value of the cdf ,

```
> phyper(q = , m = , n = , k = )
```

To graph a hypergeometric distribution as a discrete set of points, we write:

```
>x <- 0:j  
>y <- dhyper(x, m = , n = , k = )  
>plot(x, y)
```

To get quartiles, we write

qhyper (p = the given left-hand side probability, $m =, n =, k =$)

The Poisson Distribution

To select k random numbers from a Poisson distribution with mean number of occurrences λ , we write

>rpois (k , lambda = λ).

So to get 30,000 random values from a Poisson distribution with $\lambda = 1.4$, we write

>rpois (30000, lambda = 1.4)

To get the value of the pmf at x , $P(X = x)$, we write

>dpois ($x =$, lambda = λ)

To get $P(X = k), P(X = k + 1), \dots, P(X = m)$ we write

>x <- $k:m$

>dpois (x , lambda = λ)

To write this output in a tabulated format, we use the **cbind** (column bind) function and write

>w <- dpois ($k:m$, lambda = λ)

>cbind($k:m$, w)

To get the value of the cdf at x , $P(X \leq x)$, we write

>ppois ($q =$, lambda = λ)

To graph a Poisson distribution with parameter λ as a discrete set of points, we write:

>x <- $0:n$

>y <- **dpois**(x , lambda = λ)

>plot (x, y)

To get quartiles, we write

```
>qpois (p = the given left-hand side probability, lambda =  $\lambda$ )
```

So, for instance to find the x -value that has an area of 0.53 to its left in a Poisson distribution with $\lambda = 3.8$, we write

```
>qpois (p = 0.53, lambda = 3.8)
```

```
[1] 4
```

CONTINUOUS PROBABILITY DISTRIBUTIONS USING RSTUDIO

The Normal Distribution

To select k random numbers from a normal distribution with mean μ and standard deviation σ , we write

```
>rnorm (k, mean =  $\mu$ , sd =  $\sigma$ ).
```

So to get 30,000 random values from a normal distribution with mean 17 and standard deviation 3.4, we write

```
>rnorm (30000, mean = 17, sd = 3.4)
```

To get the value of the probability density function at x , we write

```
>dnorm (x, mean =  $\mu$ , sd =  $\sigma$ )
```

We can also write

```
>dnorm (k: m, mean =  $\mu$ , sd =  $\sigma$ ) to get the values of the probability density function at  $k, k + 1, \dots, m$ .
```

To get $P(X \leq x)$

```
>pnorm (x, mean =  $\mu$ , sd =  $\sigma$ )
```

To graph a normal distribution with mean μ and standard deviation σ , we write

```
>x <- rnorm (40000, mean =  $\mu$ , sd =  $\sigma$ )
```

```
>y<- dnorm (x, mean =  $\mu$ , sd =  $\sigma$ )
```

```
>plot (y~x)
```

To get a better resolution, we should use the following commands:

```
>x <- seq (lower limit, upper limit, by = 0.0005)
```

```
>y <- dnorm(x, mean =  $\mu$ , sd =  $\sigma$ )
```

```
>plot(x, y, type="l", col="..")
```

To get quartiles, we write

qnorm (the given left-hand side probability, mean = μ , sd = σ)

So, for instance to find the x-value that has an area of 0.73 to its left in a normal distribution with mean 14 and standard deviation 3, we write

```
>qnorm (p = 0.73, mean = 14, sd = 3)
```

```
[1] 15.83844
```

The Exponential Distribution

To select k random numbers from an exponential distribution with parameter λ , we write

```
>rexp (k,  $\lambda$ ).
```

So to get 30,000 random values from an exponential distribution with $\lambda = 2.7$, we write

```
>rexp (30000, 2.7)
```

To get the value of the probability density function at a point x , we write

```
>dexp (x,  $\lambda$ )
```

To get the values of the probability density function at $k, k + 1, \dots, m$ we write

```
>dexp (k:m,  $\lambda$ )
```

To get $P(X \leq x)$, we write

```
>pexp (x,  $\lambda$ )
```

To graph an exponential distribution with parameter λ , we write

```
>x <- rexp (40000,  $\lambda$ )
```

```
>y <- dexp (x,  $\lambda$ )
```

```
>plot (y~x)
```

To get a better resolution, we should use the following commands:

```
>x <- seq (lower limit, upper limit, by = 0.0005)
```

```
>y <- dexp(x,  $\lambda$ )
```

```
>plot(x, y, type="l", col="..")
```

To get quartiles, we write

```
>qexp (the given left-hand side probability,  $\lambda$ )
```

So, for instance to find the x -value that has an area of 0.53 to its left in a Poisson distribution with $\lambda = 3.8$, we write

```
>qexp (p = 0.53, 3.8)
```

The Gamma (Erlang) Distribution

To select n random numbers from a gamma distribution with parameters r (shape) and λ (scale), we write

```
>rgamma (n, r,  $\lambda$ ).
```

To get the value of the probability density function at a point x , we write

```
>dgamma (x, r,  $\lambda$ )
```

To get the values of the probability density function at $k, k + 1, \dots, m$ we write

```
>dgamma (k:m, r,  $\lambda$ )
```

To get $P(X \leq x)$, we write

```
>pgamma (x, r,  $\lambda$ )
```

To graph a gamma distribution with parameters r and λ , we write

```
>x <- rgamma (50000, r,  $\lambda$ )
```

```
>y <- dgamma (x, r,  $\lambda$ )
```

```
>plot (y~x)
```

To get a better resolution, we should use the following commands:

```
>x <- seq (lower limit, upper limit, by = 0.0005)
```

```
>y <- dgamma(x, r,  $\lambda$ )
```

```
>plot(x, y type="l", col="..")
```

To get quartiles, we write

>qgamma(the given left-hand side probability, r, λ)

So, for instance to find the x -value that has an area of 0.53 to its left in a gamma distribution with $r = 2$ and $\lambda = 4$, we write

>qgamma ($p = 0.53, 2, 4$)

The Chi Squared Distribution

To select n random numbers from a chi squared distribution with degrees of freedom k , we write

>rchisq($n, df = k$).

To get the value of the probability density function at a point x , we write

>dchisq ($x, df = k$)

To get the values of the probability density function at $m, m + 1, \dots, s$ we write

>dchisq ($m:s, df = k$)

To get $P(X \leq x)$, we write

>pchisq ($q = x, df = k$)

To graph a chi squared distribution with $df = k$, we write

>x <- rchisq (50000, $df = k$)

>y <- dchisq ($x, df = k$)

>plot ($y \sim x$)

To get a better resolution, we should use the following commands:

>x <- seq (lower limit, upper limit, by = 0.0005)

>y <- dchisq($x, df = k$)

>plot(x, y type="l", col="..")

To get quartiles, we write

>qchisq($p =$ the given left-hand side probability, $df = k$)

Here is a brief summary of some of the other frequently used distributions in statistics:

Distribution	Random Value	Density Function	Distribution Function	Quantile
Beta	rbeta	dbeta	pbeta	qbeta
Cauchy	rcauchy	dcauchy	pcauchy	qcauchy
Continuous Uniform	runif	dunif	punif	qunif
F	rf	df	pf	qf
Logistic	rlogis	dlogis	plogis	qlogis
Log-normal	rlnorm	dlnorm	plnorm	qlnorm
Multinomial	rmultinom	dmultinom	pmultinom	qmultinom
T	rt	dt	pt	qt
Weibull	rweibull	dweibull	pweibull	qweibull

SHOWING PROBABILITIES UNDER MASS (DENSITY) FUNCTIONS

In general, to show the area corresponding to $P(a < X < b)$, after getting the graph, we use the *lines* and the *polygon* function and write the following session commands:

```
>i <- (x >= a & x <= b)
```

```
>lines(x,y)
```

```
>polygon (c(a, x[i], b), c(0, y[i], 0), col = "...")
```

For example, if we want to show the area corresponding to $P(-1 < X < 1)$ (in red) under the standard normal curve (to be graphed between -4 and 4 , say, in blue), we write

```
>x <- seq (-4, 4, by = 0.0005)
```

```
>y <- dnorm(x, mean = 0, sd = 1)
```

```
>plot(x, y, type="l", col = "blue")
```

```
>i <- (x >= -1 & x <= 1)
```

```
>lines(x,y)
```

```
>polygon (c(-1, x[i], 1), c(0, y[i], 0), col = "red")
```